

Linearly Stable Self-Similar Expansion of Four and Five Point Vortices

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Abstract

Point vortices can move self-similarly, keeping their shape while the configuration grows like $\sqrt{t}$ and turns. For three vortices such expanding configurations are stable, and Zbarsky used this to show that vortex patches placed at the three vortices stay confined for all time; he noted that the extension to four or more vortices needs a self-similarly expanding configuration with good stability, and that none was known. We give two. For the circulations $(-1, -5/2, -1/9, 4/5)$ and $(-1, 3/7, 7/8, -9/7, -47/35)$ there are self-similarly expanding configurations of four and five Euler point vortices whose linearization in similarity variables has, besides the double eigenvalue 0 of the rotation and of the one-parameter family of such configurations, only eigenvalues with real part -1 or -2 : every perturbation transverse to the rotation and to the family decays at least like the inverse of the size of the configuration. The proof is computer-assisted: the configurations are enclosed by the Krawczyk method in ball arithmetic, and the spectrum is controlled through an exact lemma, which shows that six eigenvalues are forced by the symmetries and that the others come in pairs $k, 2 - k$, together with enclosures of traces of powers of the Jacobian. The property is open, so such configurations form open sets; in a random sample about one four-vortex collapse in seven and one five-vortex collapse in seventeen reverses into a stable expansion. We do not address whether this linear stability suffices for Zbarsky's confinement argument.

Keywords: point vortices; self-similar motion; linear stability; computer-assisted proof; interval arithmetic.

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1 Introduction

N point vortices with circulations Γ_j at positions $z_j \in \mathbb{C}$ move by

$$\dot{\bar{z}}_j = \frac{1}{2\pi i} \sum_{k \neq j} \frac{\Gamma_k}{z_j - z_k}. \quad (1)$$

A motion is self-similar when the configuration keeps its shape while it grows or shrinks and turns: $\dot{z}_j = \kappa(z_j - z_c)$ with one $\kappa \in \mathbb{C}$, where $z_c = \sum_j \Gamma_j z_j / \sum_j \Gamma_j$. Then the size of the configuration grows or shrinks like $|1 + 2t \operatorname{Re} \kappa|^{1/2}$, the configuration expands when $\operatorname{Re} \kappa > 0$ and collapses in finite time when $\operatorname{Re} \kappa < 0$ [7, 9, 1, 2, 6]. Reversing every circulation reverses time, so each collapse becomes an expansion of the same shape, and conversely.

Zbarsky [11] proved that for a generic self-similarly expanding configuration of three vortices one may replace each vortex by a small patch of vorticity and the patches stay confined for all time, their radius growing more slowly than their distance. The restriction to three vortices comes from the stability analysis. In his words, if one found stably growing systems of four or more vortices the confinement result would most likely carry over with little modification, but one needs sufficiently good stability results, and orbital linear stability is not enough [11, pp. 4–5 of the arXiv version]; the same point recurs in [12, after Cor. 16]. For three vortices, stability of the expanding family goes back to Tavantzis and Ting [10]; Iwayama and Yajima studied the linear stability of the self-similar motions of three vortices in the α -models [5].

We show that stable expansions of more vortices exist. Section 2 sets up similarity variables, in which a self-similar motion is a fixed point, and proves a lemma on the structure of the spectrum of the linearization (Lemma 1): six eigenvalues are forced by the rotation, the scaling, the translations and the one-parameter family of self-similar configurations with the given circulations, and the others come in pairs $k, 2 - k$. For three vortices nothing else is left. For four vortices one pair is left, and a single number decides its stability; for five, two pairs. Section 3 states and proves, with a computer-assisted proof, that there are configurations of four and of five vortices whose reversal expands with every other exponent of real part 1 (Theorems 1 and 2, Corollary 1). Section 4 gives numerical illustrations, controls and a sample of how common such configurations are, and Section 5 discusses what is and is not shown.

2 Similarity variables and the structure of the spectrum

2.1 Similarity variables

Let $V(\zeta) = (V_1, \dots, V_N)$ be the right side of (1), $V_j = (i/2\pi) \sum_{k \neq j} \Gamma_k / (\zeta_j - \zeta_k)$. It is homogeneous of degree -1 , $V(\lambda\zeta) = \lambda^{-1}V(\zeta)$ for real $\lambda > 0$, and equivariant under rotations, $V(e^{i\phi}\zeta) = e^{i\phi}V(\zeta)$. For $\kappa = a + i\omega$ put $z_j(t) = \rho(t)e^{i\omega\tau}\zeta_j(\tau)$ with $\rho^2 = 1 + 2at$ and $d\tau/dt = \rho^{-2}$; then (1) becomes

$$\frac{d\zeta}{d\tau} = V(\zeta) - \kappa\zeta, \quad (2)$$

and a self-similar motion with this κ and $z_c = 0$ is a fixed point of (2). We fix the scale of time by $2\pi a = -1$ for a collapse, write $2\pi\kappa = -1 + ib$, and work with $E(\zeta) = 2\pi(V(\zeta) - \kappa\zeta) = 2\pi V(\zeta) + (1 - ib)\zeta$ and the rescaled time $s = \tau/2\pi$; for a collapse, $s = -\ln \rho$. The winding number of the collapse is $P = |b|/2$ [4]. Let DE be the real $2N \times 2N$ Jacobian of E at a zero ζ^* . A perturbation along an eigenvector with eigenvalue k behaves like $e^{ks} = \rho^{-k}$ relative to the size ρ of the configuration. Writing $\delta\zeta = u + iv$, one finds

$$DE = \begin{pmatrix} A_r + I & A_i + bI \\ A_i - bI & -A_r + I \end{pmatrix}, \quad A_{jk} = \frac{i\Gamma_k}{(\zeta_j - \zeta_k)^2} \ (k \neq j), \quad A_{jj} = -\sum_{k \neq j} A_{jk}, \quad (3)$$

with $A = A_r + iA_i$ evaluated at ζ^* .

Time reversal. If ζ^* is a collapse for the circulations Γ , it is a fixed point of the similarity dynamics of the circulations $-\Gamma$ with κ replaced by $-\kappa$, an expansion with the same winding, and the Jacobian of its similarity dynamics is $-DE$; the similarity time is then $s = \ln \rho$. So a mode with eigenvalue k of DE behaves in the expansion like ρ^{-k} relative to its size, and it decays as the configuration grows exactly when $\text{Re } k > 0$.

2.2 The structure of the spectrum

Lemma 1. *Let ζ^* be a zero of E , with $b \neq 0$, all $\Gamma_j \neq 0$, $\sum_j \Gamma_j \neq 0$ and the ζ_j^* distinct, and suppose that near ζ^* the zeros of E with the same circulations, up to rotation, form a smooth curve through ζ^* that is transverse to the rotations. Then the characteristic polynomial p of DE satisfies $p(\lambda) = p(2 - \lambda)$ and is divisible by*

$$\lambda^2(\lambda - 2)^2((\lambda - 1)^2 + b^2),$$

and the quotient q , of degree $2N - 6$, also satisfies $q(\lambda) = q(2 - \lambda)$. In particular the $2N - 6$ remaining eigenvalues, counted with multiplicity, come in pairs $k, 2 - k$.

Proof. Rotation. $E(e^{i\phi}\zeta) = e^{i\phi}E(\zeta)$, so differentiating at $\phi = 0$ gives $DE(i\zeta^*) = iE(\zeta^*) = 0$.

Scaling. $E(\lambda\zeta) = 2\pi V(\zeta)/\lambda + (1 - ib)\lambda\zeta$; at $\lambda = 1$ the derivative is $-2\pi V(\zeta^*) + (1 - ib)\zeta^* = 2(1 - ib)\zeta^*$, since $2\pi V(\zeta^*) = -(1 - ib)\zeta^*$. So $DE\zeta^* = 2\zeta^* - 2bi\zeta^*$, and on the span of ζ^* and $i\zeta^*$ the map DE is triangular with eigenvalues 2 and 0.

Translations. V is invariant under $\zeta \mapsto \zeta + c(1, \dots, 1)$, so DE acts on $\mathbf{1} = (1, \dots, 1)$ and $i\mathbf{1}$ as multiplication by $1 - ib$: $DE\mathbf{1} = \mathbf{1} - bi\mathbf{1}$, $DE(i\mathbf{1}) = i\mathbf{1} + b\mathbf{1}$, with eigenvalues $1 \pm ib$.

The family. Along the curve of zeros, $E(\zeta(\sigma); b(\sigma)) = 0$, and $\partial_b E = -i\zeta$. Differentiating, $DE\zeta'(0) = b'(0)i\zeta^*$, so $DE^2\zeta'(0) = 0$. As $\zeta'(0)$ is not a real multiple of $i\zeta^*$, the generalized kernel of DE has dimension at least 2: the algebraic multiplicity of 0 is at least 2.

Pairing. With $H = -\frac{1}{2}\sum_{j \neq k} \Gamma_j \Gamma_k \ln |\zeta_j - \zeta_k|$ and $I = \sum_j \Gamma_j |\zeta_j|^2$, the field $\zeta \mapsto 2\pi V(\zeta) - ib\zeta$ is the Hamiltonian vector field of $H + (b/2)I$ for the symplectic form $\sum_j \Gamma_j dx_j \wedge dy_j$, which is nondegenerate because every $\Gamma_j \neq 0$ (Kirchhoff's form of (1) and the rotations generated by I , up to the sign convention of the form). The linearization of a Hamiltonian vector field at any point is an infinitesimally symplectic matrix, whose characteristic polynomial is even. Since DE is this linearization plus the identity, $p(\lambda) = r(\lambda - 1)$ with r even, that is $p(\lambda) = p(2 - \lambda)$, with multiplicities.

Conclusion. By pairing, the multiplicity of 2 equals that of 0, at least 2. Since $b \neq 0$, the numbers 0, 2, $1 + ib$, $1 - ib$ are distinct, so p is divisible by the displayed product. That product is invariant under $\lambda \mapsto 2 - \lambda$, hence so is the quotient. $\square$

For three vortices $2N - 6 = 0$: in similarity variables every exponent of a three-vortex expansion is forced by the symmetries and the family, which is the linear shadow of the classical stability of the expanding three-vortex family [10]. For four vortices one pair $k, 2 - k$ remains, and with $c = k(2 - k)$ the sum of squares of all eigenvalues gives

$$\text{tr}(DE^2) = (0 + 0 + 4 + 4 + 2(1 - b^2)) + (4 - 2c), \quad c = 3 + 3b^2 - \text{tr}(A\bar{A}), \quad (4)$$

using $\text{tr}(DE^2) = 2\text{tr}(A\bar{A}) + 2N(1 - b^2)$, which follows from (3). Both remaining exponents have positive real part exactly when $c > 0$; when $c > 1$ they are $1 \pm i\sqrt{c - 1}$. For five vortices two pairs remain; with $u = (k - 1)^2$ they are the roots u_1, u_2 of $u^2 - su + p$, where $s = u_1 + u_2$ and $u_1^2 + u_2^2$ follow from $\text{tr}((DE - I)^2)$ and $\text{tr}((DE - I)^4)$ after the contributions $4 - 2b^2$ and $4 + 2b^4$ of the six forced eigenvalues are removed. Every remaining exponent has positive real part exactly when every root u satisfies $|\text{Re} \sqrt{u}| < 1$; in particular it suffices that both roots are real and less than 1.

3 Results

Theorem 1 (Four vortices). *For the circulations $\Gamma = (1, 5/2, 1/9, -4/5)$ there is a self-similar collapse of four Euler point vortices with $2\pi\kappa = -1 + ib$, winding*

$$P = b/2 = 2.5258464741344909107270\dots,$$

and z_4 with real part $8/25$ in the gauge $z_1 \in \mathbb{R}$, $z_c = 0$, whose Jacobian DE has the six eigenvalues $0, 0, 2, 2, 1 \pm ib$ of Lemma 1 and the pair $1 \pm i\omega$ with $\omega = 4.4588620962481 \dots$; here $c = 20.881451193358 \dots$ in (4).

Theorem 2 (Five vortices). *For the circulations $\Gamma = (1, -3/7, -7/8, 9/7, 47/35)$ there is a self-similar collapse of five Euler point vortices with winding $P = 2.40234703541713841 \dots$ and z_2 with real part $3/5$ in the same gauge, whose Jacobian DE has the six eigenvalues of Lemma 1 and the four eigenvalues $1 \pm 3.4538555 \dots i$ and $1 \pm 3.1885675 \dots i$.*

Corollary 1. *For the circulations $(-1, -5/2, -1/9, 4/5)$ and $(-1, 3/7, 7/8, -9/7, -47/35)$ the configurations of Theorems 1 and 2 expand self-similarly, growing like $\sqrt{t}$. In similarity variables the linearization at them has the double eigenvalue 0, whose generalized eigenspace consists of the rotation and the tangent of the one-parameter family of expanding configurations with the same circulations, and every other eigenvalue has real part -1 or -2 . Every perturbation transverse to the rotation and to the family therefore decays, relative to the size ρ of the configuration, at least like ρ^{-1} , that is like $t^{-1/2}$.*

Proof of the theorems. The program `verify_stable_expansion.py` (Section 4) works in the gauge of [4, Sect. 8]: $\Gamma_1 = 1$, z_1 real, $\sum_k \Gamma_k/(z_j - z_k) = \lambda \bar{z}_j$ with $\lambda = 2P - i$, which is $E = 0$ with $b = 2P$ and $z_c = 0$. For four vortices it fixes $x_4 = 8/25$, $\Gamma_2 = 5/2$ and $\Gamma_3 = 1/9$ and proves by the Krawczyk test, in the ball arithmetic of FLINT/Arb, that the 8 equations have exactly one zero in a box of radius 10^{-4} around the numerical solution; on the box the circulations are nonzero, the total circulation is nonzero, the vortices are distinct and away from the origin, and the necessary conditions $\sum_j \Gamma_j z_j = 0$, $\sum_{j < k} \Gamma_j \Gamma_k = 0$ and $\sum_j \Gamma_j |z_j|^2 = 0$ are enclosed with 0. The second of these forces $\Gamma_4 = -4/5$. The Jacobian of the square system is invertible on the box, so by the implicit function theorem the zeros with Γ_2, Γ_3 fixed form a smooth curve parametrized by x_4 ; along it Γ_4 stays $-4/5$, and its tangent v has $\text{Im } v_1 = 0$ and v_4 with real part 1, while $i\zeta^*$ has $\text{Im}(i\zeta_1^*) = x_1 \neq 0$, so v is not a real multiple of $i\zeta^*$. The hypotheses of Lemma 1 hold, and the program also checks on the box that the vectors of its proof satisfy their identities. It then encloses $c = 3 + 3b^2 - \text{tr}(A\bar{A})$ by $20.8814511933587 \pm 1.4 \cdot 10^{-14}$, so $c > 1$ and the remaining pair is $1 \pm i\sqrt{c-1}$; a second enclosure from $\text{tr}((DE - I)^2)$ agrees. For five vortices it fixes $x_2 = 3/5$ and $\Gamma_2, \Gamma_3, \Gamma_4$, proves existence the same way, finds $\Gamma_5 = 47/35$, checks the hypotheses of the lemma, and encloses s and p ; the discriminant $s^2 - 4p$ is positive and both roots, $u_1 = -11.929118 \dots$ and $u_2 = -10.166963 \dots$, are negative, so the four remaining eigenvalues are $1 \pm i\sqrt{-u_1}$ and $1 \pm i\sqrt{-u_2}$. $\square$

Proof of the corollary. By the time reversal of Section 2 the configurations expand for the negated circulations, with Jacobian $-DE$ in similarity variables and similarity time $s = \ln \rho$. The eigenvalues of $-DE$ are $0, 0, -2, -2, -1 \mp ib$ and, by the theorems, $-1 \mp i\omega$ or $-1 \mp i\sqrt{-u_{1,2}}$. By the proof of Lemma 1 the generalized eigenspace of 0 is spanned by $i\zeta^*$ and the tangent of the family. On an invariant complement every solution of the linearized equation decays like $e^{-s} = \rho^{-1}$ at least, and ρ^2 grows linearly in t . $\square$

Remark 1 (Open sets). The conclusions are open conditions: c , and the roots u , depend continuously on the configuration and the circulations along the collapse set, which near these points is a smooth manifold of dimension $N - 1$ (modulo translations, rotations and the scalings of lengths and circulations) [4, Sect. 7]. So every collapse near those of the theorems, with nearby circulations, reverses into a linearly stable expansion as well; the stable expansions of four and five vortices form open sets, not isolated examples.

program	what it checks	checks
<code>verify_stable_expansion.py</code>	Theorems 1, 2, the hypotheses of Lemma 1 on the boxes, controls, integrations	46
<code>survey_expansions.py</code>	the random sample of Section 4 (numerical)	

Table 1: The programs. Each prints every check and stops with an error if one fails; the output is in `data/`.

4 Numerical illustration, controls and a sample

Controls. The same program certifies two controls. A four-vortex collapse with circulations $(1, -1003/1000, 13/10, \Gamma_4)$ has $c = -47.7346\dots < 0$: its remaining pair is real, $k < 0$, and its reversal is unstable. A three-vortex collapse has $\text{tr}((DE - I)^2) = 4 - 2b^2$ to within its enclosure, the contribution of the six forced eigenvalues alone, as Lemma 1 requires for $N = 3$.

Direct integration (binary64, DOP853, relative tolerance 10^{-12} ; not part of the proof). The four-vortex expansion of Corollary 1, started from the midpoint of the certified box, keeps its shape to $4 \cdot 10^{-15}$ while its size grows 56-fold; a random perturbation of relative size 10^{-5} stays below $2 \cdot 10^{-4}$, settling on a nearby member of the family, the neutral direction. The reversal of the unstable control, perturbed by 10^{-8} , departs from its shape by $1.5 \cdot 10^{-3}$ while its size grows eightfold. A self-similar collapse, by contrast, departs from self-similarity in binary64 under round-off alone [8, 3].

How common. Newton’s method from random starts, in binary64 with fixed seeds, gave 342 distinct four-vortex collapses, of which 52 have $c > 0$, and 543 five-vortex collapses, of which 32 have every remaining exponent with positive real part (`survey_expansions.py`). This is a sample of what a naive search finds, not a measure on the collapse set. In most five-vortex cases both remaining pairs lie on $\text{Re } k = 1$, as in Theorem 2; a few have a quadruple $1 \pm x \pm iy$ with $0 < x < 1$.

5 Discussion

The theorems give self-similarly expanding configurations of four and five vortices that are linearly asymptotically stable modulo the rotation and the family of expansions with the same circulations, with a spectral gap: every transverse perturbation decays like a negative power of t . This is more than orbital linear stability, which Zbarsky notes is not enough for his argument [11], but it is still a linear statement. We have not proved nonlinear stability, although a spectral gap of this kind is the usual starting point for it, and we have not examined whether the confinement argument of [11, 12] goes through for these configurations. The circulations of the theorems are not symmetric and the configurations have no symmetry; by Remark 1 they are not special.

The examples are Euler vortices. The α -models, where a vortex induces the velocity $\Gamma r^{-\alpha-1}/(2\pi)$, have the same structure after the obvious changes of homogeneity, and we have not treated them. Nor have we asked which fraction of the collapse set reverses into stable expansions, or whether stable expansions exist for every $N \geq 4$; the sample of Section 4 suggests that they are common for $N = 4$ and 5.

Data availability. The programs `verify_stable_expansion.py` and `survey_expansions.py`, with the modules `certify_ball_ad.py` and `certify_pipeline.py` of [4], their inputs and their output will be in the repository <https://github.com/ChaseHendrick/stable-expansion>: the programs in `code/` and their inputs and output in `data/`.

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